Ismail R. Mohamed

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Research Interests

Hardware–software co-design for trustworthy embodied AI: world-model reliability, runtime integrity monitoring of learned perception, AI hardware, neuromorphic computing, and real-time digital twins for scientific computing.

Education

The University of Texas at Dallas — Richardson, TX, USA
Ph.D. in Computer Engineering; Advisor: Prof. Mohamed Ibrahim (EACAS Lab) — *Aug. 2025 – Present*

The American University in Cairo — New Cairo, Egypt
B.Sc. in Electronics and Communication Engineering — *Aug. 2020 – Jun. 2025*

Portland State University — Portland, OR, USA
Exchange Semester, Department of Electrical and Computer Engineering — *Jan. 2024 – Jun. 2024*

- Coursework: Verilog/SystemVerilog, Low-Power IC Design Techniques, ASIC Design

Publications

[C2] **Ismail R. Mohamed**, Mohamed Ibrahim. "PRISM: Few-Shot Runtime Integrity Monitoring of Learned Perception via Corruption Subspace Families." *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026. To appear. **Best Paper Candidate.**

[C1] Hammad Omar, **Ismail R. Mohamed**, Mohamed Ibrahim. "ENACT: Ensemble Neural Fields for Reactive Robot Control." *ACM/IEEE Design Automation Conference (DAC)*, 2026. doi:10.1145/3770743.3804407

Under Review

[S1] **Ismail R. Mohamed**, et al. "CoTwin: Numerical–Hardware Co-Design for Real-Time Edge Digital Twins Under Changing Physics." Under review at *DAC*, 2027.

Honors & Awards

Best Paper Candidate, IEEE/ACM ICCAD (for PRISM) — 2026

Exchange Scholarship, Portland State University exchange semester — 2024

UAMA Merit Scholarship, The American University in Cairo — ranked 5th nationally in Mathematics — 2020

Research Experience

Graduate Research Assistant — Aug. 2025 – Present
EACAS Lab, The University of Texas at Dallas — Advisor: Prof. Mohamed Ibrahim — *Richardson, TX*

- **PRISM (ICCAD'26, Best Paper Candidate):** Led a few-shot runtime integrity monitor that detects silent degradation of learned perception under sensory corruption. Showed that diverse corruptions concentrate into a small set of subspace families, enabling a scorer that generalizes to unseen corruptions at sub-millisecond overhead across safety-critical streaming domains.
- **CoTwin (under review):** Leading a numerical–hardware co-design for real-time digital twins on edge hardware under changing physics.
- **ENACT (DAC'26):** Contributed to ensemble neural fields for reactive robot control.

Undergraduate Research Assistant — AI Acceleration on SoC FPGA — Jun. 2024 – Jun. 2025
The American University in Cairo — Advisor: Prof. Ahmed Abu Ouf — *New Cairo, Egypt*

- Accelerated 3D convolutional neural networks for LiDAR-based perception by partitioning the pipeline between the processing system (feature extraction) and the FPGA fabric (network inference).
- Responsible for algorithm development, system design, and performance analysis for real-time inference.

Hardware Research Intern — Jul. 2023 – Aug. 2023
Center of Nanoelectronics and Devices (CND), AUC — Mentor: Dr. Yehea Ismail — *New Cairo, Egypt*

Teaching Experience

Teaching Assistant — CE 6320: Applied Data Structures and Algorithms — Fall 2026
The University of Texas at Dallas — *Richardson, TX*

Teaching Assistant — Circuits — Jan. 2023 – Jan. 2024
The American University in Cairo — *New Cairo, Egypt*

Professional Service

Reviewer, ACM/IEEE Design Automation Conference (DAC) 2026

Reviewer, Asia and South Pacific Design Automation Conference (ASP-DAC) 2027

Leadership & Activities

Chair, IEEE AUC Student Branch — led a 50-member branch and set its strategic direction Jun. 2022 – Jun. 2023

Student Ambassador & University Representative, The American University in Cairo Oct. 2021 – Jun. 2025

Electronics Co-Lead, AUC Robotics Club ROV Team — designed PCBs for ROV control

Professional Training

Siemens Calibre D2S Diploma Sep. 2024 – Nov. 2024
Siemens EDA *New Cairo, Egypt*

- VLSI design flow, fabrication processes, multi-patterning, physical verification (DRC/LVS/PEX), design for manufacturability, parasitic extraction, and 3D stacking.

Digital IC Design Training Aug. 2023 – Jan. 2024
Ministry of Communications, in collaboration with AUC (CND) *New Cairo, Egypt*

- Full-custom CMOS design flow, standard-cell library development, SPICE-based circuit analysis, and FPGA implementation of digital circuits and FSMs.

Technical Skills

Programming: Python, C/C++
Machine Learning: PyTorch
Hardware Design: Verilog, SystemVerilog, FPGA and SoC FPGA design
Tools: AMD/Xilinx Vivado, Intel Quartus, Git, LaTeX